

SURG70004 Image Guided Intervention

IGI Coursework 2 – Medical Scene Understanding

Due date Friday 24th January 2025 16:00 (via Blackboard)

During this exercise you will investigate the 3D reconstruction of a surgical scene and you will focus on the detection of salient features on the tissue surface and the estimation of stereo feature correspondences. For this purpose, in vivo data from the SCARED dataset [1] will be used with known camera calibration parameters. In addition, you will explore volumetric visualisation methods and estimate volumetric renderings using the 3D scan of a zebrafish.

The questions are provided in black font. Please use blue for your answers in the spaces indicated.

Task 1 – Epipolar Geometry (20%)

Two cameras are placed at positions $CL=[-5,0,0]$ and $CR=[5,0,0]$ and both face in direction $[0, 0, 1]$.

- i) Two feature points are identified on one image plane as $[5,0]$ and $[-4,0]$, the origin being at the centre of the image plane. What are the epipolar lines corresponding to these two points?
- ii) A point on an object has 3D position $P_i=[X_i, Y_i, Z_i]$. Derive an expression for the epipolar plane where point P_i lies.
- iii) If the image plane of projection lies in the plane $Z=7$, extract the equation of the epipolar line that appear in the left image.
- iv) The above cameras are focused on the abdomen of a patient lying naked on the operating bed. Propose a method of recovering the respiratory rate of the patient. You can use any additional equipment in the operating theatre.

[Answer here \(expand as necessary\):](#)

Task 2 – OR Coordinate Frames (20%)

The camera shown in Figure 1, has its $[x_c, y_c, z_c]$ axes aligned with the world $[y_w, z_w, x_w]$ axes, respectively. The origin of the world coordinate frame is at $[0, -d, 6d]$ in the camera coordinate frame.

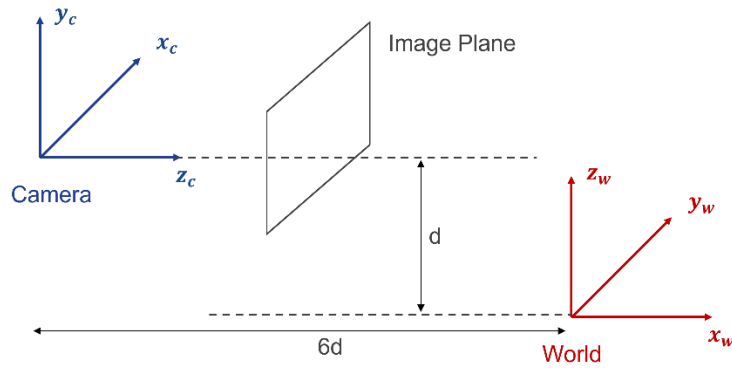


Figure 1: Coordinate frame transformations.

- Estimate the transformation $[R|T]$ between the camera and the world coordinate frames such that $X_c = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} X_w$.
- For the above camera, the focal length along the x and the y axis is $\alpha_x = 800$ and $\alpha_y = 720$, the principal point is at $(x_0, y_0) = (350, 250)$ and the skew is zero. Derive the coordinates of the projection of the origin of the world coordinate frame on the image plane.

[Answer here \(expand as necessary\):](#)

Task 3 – 3D Scene Reconstruction (40%)

The following are example frames from a video sequence.

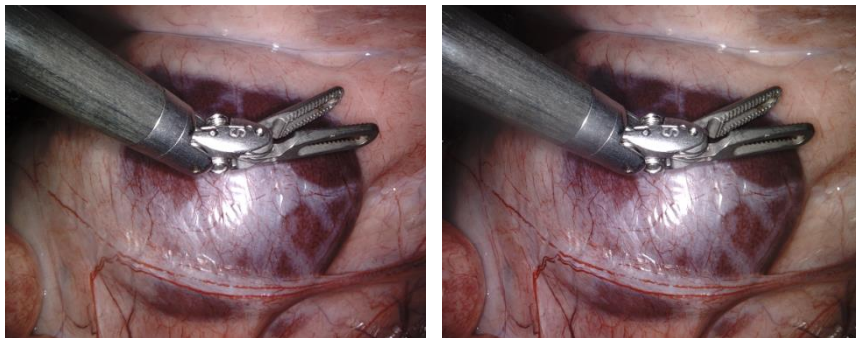


Figure: Stereo images included in the dataset. (Left) Left camera frame (Right) Right camera frame.

Use a programming environment of your choice and the provided stereo images to:

- Detect salient features and plot the detected features on the provided pair of frames.
- Find corresponding features between the two frames. Create a composite image (e.g. centred overlay image) from the two frames and demonstrate the matched points.
- Use the matched features to estimate the fundamental matrix between the two images. Now estimate the fundamental matrix using the extrinsic and intrinsic camera parameters.

Compare the estimated fundamental matrices and explain any possible disagreement between the two methods. Which method is more accurate? Justify your answer.

- d. Find the correctly matched points that meet the epipolar constraint and illustrate these matches. Briefly explain how these matches have been identified.
- e. Estimate the diameter of the shaft of the tool shown in the stereo images. (hint: you can establish the disparity map between the left and right camera images or you can apply 3D surface reconstruction)

For each of the above tasks, provide a brief explanation about the approach you followed to estimate your results.

[Answer here \(expand as necessary\):](#)

Task 4 - Visualisation (20%)

- a. Draw a set of transfer functions that can be used to create a rendering of a CT dataset of the abdomen. Assume that the signal in CT units from bone is >80 , that from liver is 25-50 and that the signal from fat is 10-25. The tissues appear semi-realistic, i.e. bone should be white and opaque, liver should be dark red and opaque, fat should be white and semi-transparent.
- b. In the provided coursework datasets, you can find the optical projection tomography volumetric data of a zebrafish. Implement the MIP method and apply it to project the volumetric data along the 3 orthogonal directions (x, y and z-axis). Display each projection plane of your rendering.

[Answer here \(expand as necessary\):](#)

References

[1] Allan M, McLeod AJ, Wang CC, Rosenthal J, Hu Z, Gard N, Eisert P, Fu KX, Zeffiro T, Xia W, et al. 2021. Stereo correspondence and reconstruction of endoscopic data challenge. CoRR. abs/2101.01133